

WEEK 7:

Financial Ratio

Relationship between two financial statements – sales vs assets, net profits vs equity, debt vs equity

Liquidity Ratios:

- Measures cash ability of a firm – assessing if a firm has sufficient short term resources to pay its short term obligations

$$\text{liquidity ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$


- Current ratio** – compares total current assets to current liabilities

Standard ratio: 1.33:1

- Quick ratio:** removes inventory from assets – used for short term obligations

$$\text{Ratio} = \frac{\text{current assets} - \text{inventory}}{\text{current liabilities}}$$

Solvency Ratios:

- Debt-equity ratio:** compares long term debt to total equity (owner capital + reserves)
 - Standard ratio: 2:1

$$\text{Ratio} = \frac{\text{long term debt}}{\text{total equity}}$$

- Interest earned ratio:** firm's capacity to pay interest obligations

$$\text{Ratio} = \frac{\text{Profit after Tax} + \text{Interest on long term debt} + \text{Non cash charges}}{\text{Interest on long term debt}}$$

- Non-cash charges can be depreciation
- Standard ratio:** 2 to 3 times

Efficiency/ Turnover ratio:

Debtors' turnover ratio:

- Evaluates operating efficiency of a firm – how quickly it converts its current assets into sales and cash.

$$\text{Ratio} = \frac{\text{total credit sales}}{\text{debtors}}$$

- Measures how quickly credit sales are collected
Higher ratio – better

Inventory turnover ratio:

- How many times a company has sold and replaced inventory during a specific period.

$$\text{Ratio} = \frac{\text{total sales}}{\text{avg inventory}}$$

- Helps identify if companies are overstocking/items are sitting on the shelf for too long

Total asset turnover ratio

- How efficiently is a company using its assets

$$\text{Ratio} = \frac{\text{total sales}}{\text{total assets}}$$

- **Higher ratio: better** – company is generating more revenue per rupee of total assets

Profitability ratios:

- Allows us to measure the ability of the firm to earn a return on sales/total assets/capital

Profit margin:

- How much the company earns on sales

$$\left\{ \frac{\text{Profit after tax}}{\text{Sales}} \right\}$$

- High profit margin – more efficient company within the same industry

Return on assets (ROA)

- How profitable a company is relative to its total assets

$$\frac{\text{Profit after Tax}}{\text{Total Assets}}$$

- Higher ROA – better

Return on equity (ROE)

- How profitable a company is relative to its total equity

$$\frac{\text{Profit after Tax}}{\text{Shareholder's equity}}$$

→ got from balance sheet

- Key performance indicator**
- Also used to verify if the return is better by taking this risk than leaving money in the bank which is risk free

Return on Capital Employed (ROCE)

$$\frac{\text{Earnings before Interest \& Tax} \rightarrow \text{EBIT}}{\text{Equity + long term debt} \rightarrow \text{Total capital employed}}$$

Market Valuation Ratios

Relate to company's market price per share to a performance metric like earnings per share

Earnings per share (EPS)

$$\frac{\text{Profit after Tax}}{\text{Total number of outstanding equity shares}}$$

→ equity shares in the market
→ got from balance sheet

- Used to determine their per share return

Price to earning ratio (P/E)

$$\frac{\text{Market price per share}}{\text{Earning per share (EPS)}}$$

- High ratio – stock price is high related to earnings – overvalued / investors expect high growth in future

Dividend yield ratio

$$\frac{\text{dividend per share}}{\text{market price per share}}$$

- Return on the stock vs a bond or savings account